

Junior Seminar Independent Work: Introduction to the Torelli group

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1 Previous Theorem and Propositions

This section will briefly list previous Propositions that are referenced throughout this paper.

Proposition 2.3. The natural map

$$Mod(S_3) \rightarrow \Sigma_3$$

given by the action of $Mod(S_{0,3})$ on the set of marked points of $S_{0,3}$ is an isomorphism.

Proposition 3.1. Let a be the isotopy class of a simple closed curve α in a surface S . If α is not homotopic to a point or a puncture of S , then the Dehn twist T_a is a nontrivial element of $Mod(S)$.

Proposition 6.3. Let a and b be isotopy classes of oriented simple closed curves in S_g . For any $k \geq 0$, we have

$$\Psi(T_b^k)([a]) = [a] + k \cdot \hat{i}(a, b)[b]$$

2 Definitions and Motivation

The Mapping Class Group, $MCG(S)$, of some surface S is defined as follows:

Definition 2.1 (Mapping Class Group). $MCG(S)$ is the group of isotopy classes of orientation preserving diffeomorphisms of S that they restrict to identity on ∂S .

$$MCG(S) = \text{Diff}^+(S, \partial S) / \text{Diff}_0(S, \partial S)$$

MCGs consists of elements acting non-trivially on $H_1(S, \mathbb{Z})$ and those acting trivially on $H_1(S, \mathbb{Z})$. Recall that we have undergone similar explorations of these two categories in previous sections. For instance, for the sphere with three punctures, $S_{0,3}$, we show that elements of $MCG(S_{0,3}) \cong \Sigma_3$ acting trivially on $H_1(S_{0,3})$, that is, fixes the 3 punctures, is the trivial mapping class (Proposition 2.3). In another instance, we show that Dehn twists on non-separating simply closed curves are non-trivial elements of $MCG(S)$ (Proposition 3.1).

The purpose of Torelli groups is to investigate the structure of the **trivial** elements of MCGs.

Definition 2.2 (Torelli group). $\mathcal{I}(S_g)$ is a normal subgroup of $MCG(S_g)$ satisfying the exact sequence

$$1 \rightarrow \mathcal{I}(S_g) \xrightarrow{\triangleleft} MCG(S_g) \xrightarrow{\Psi} Sp(2g, \mathbb{Z}) \rightarrow 1,$$

that is, $\mathcal{I}(S_g)$ are the elements of $MCG(S_g)$ that act trivially on $H_1(S_g)$

Note that the surjective map

$$\Psi : MCG(S_g) \rightarrow Sp(2g, \mathbb{Z})$$

is the symplectic representation of the MCG. The symplectic representation captures the non-trivial elements of MCG well as it arises from the map

$$\Psi_0 : MCG(S_g) \rightarrow Aut(H_1(S_g; \mathbb{Z}))$$

. Thus we can think of the Torelli group as encoding the structure that is otherwise not captured by the symplectic representation.

3 Examples of Elements of a Torelli group

Below are a few ways elements of a Torelli group arise.

1. Dehn twists about separating simple closed curves.

For any S_g , there exists a basis for $H_1(S_g)$ s.t. each element is disjoint from some separating simple closed curve γ . Therefore if a Dehn Twist T_γ fixes the separating curve, it would also fix the corresponding homology classes.

To make this explicit and technical, Proposition 6.3 states that any Dehn Twist can be written as follows

$$T_\gamma(x) = x + \hat{i}([\gamma], x)[\gamma],$$

where $x \in H_1(S_g, \mathbb{Z})$ and $\hat{i}$ measures the number of intersections between x and $[\gamma]$. γ separating and said choice of basis for $H_1(S_g, \mathbb{Z})$ implies that $\hat{i}([\gamma], x) = 0$, thus $T_\gamma(x) = x$.

This also leads to the subgroup of the Torelli group $\mathcal{K}(S_g)$ generated by Dehn twists about separating simple closed curves.

2. Bounding Pair Maps

The notion of Bounding Pair Maps is defined as follows.

Definition 3.1 (Homologous). Two curves a, b are homologous if they bound a surface.

Definition 3.2 (Bounding Pair). Two curves a, b is a Bounding Pair if it is disjoint, homologous, (individually) non-separating simple closed curves. See Fig. 1 for a counterexample and an example.

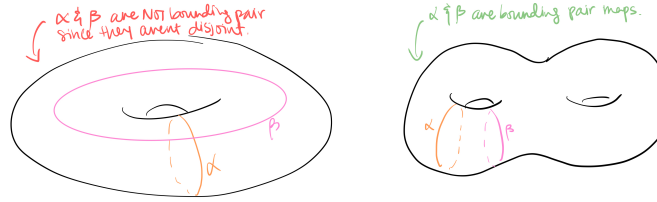


Figure 1: Counterexample and Example of Bounding Pair

Definition 3.3 (Bounding Pair Map). A Bounding Pair Map is a mapping class given by $T_a T_b^{-1}$, that is, the Dehn twist and inverse Dehn twist about a Bounding Pair a, b .

Note that by Proposition 6.3, since a, b are homologous, T_a and T_b has the same symplectic representation in $Sp(2g, \mathbb{Z})$, therefore, $T_a T_b^{-1}$ is trivial and is an element of $\mathcal{I}(S_g)$.

We also briefly mention "fake" Bounding Pair Maps. In the previous discussion, we don't actually use the disjoint condition of Def 3.2. Then we get a special case of the above by imposing a disjoint condition. For curves c, d with $\hat{i}(c, d) = 0$, we have that

$$T_c T_d T_c^{-1} T_d^{-1} = T_c T_{T_d(c)}^{-1},$$

where c and $T_d(c)$ are homologous.

3. Point pushes

Previous discussions on the Birman Exact Sequence introduced point pushing homomorphism,

$$\text{Push} : \pi_1(S_g) \rightarrow MCG(S_g)$$

where $\pi_1(S_g)$ is the fundamental group of S_g . This map takes elements of the fundamental group to bounding pair maps in $MCG(S_g)$.

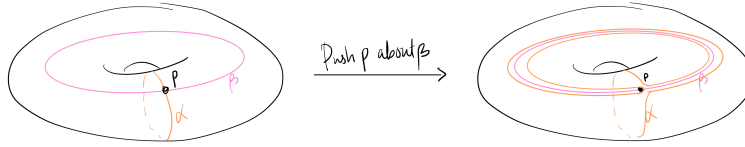


Figure 2: Example of Point Pushing Map

In particular, we have, $\text{Push}(\pi_1(S_g)) \subseteq \mathcal{I}(S_g)$.

This concludes our discussion on examples of Torelli group elements. In the next section, we investigate applications of the Torelli group on surfaces we have seen prior.

4 Torelli group of the Torus (T^2)

On the torus T^2 , it follows from Theorem 2.5 that $MCG(T^2) \cong SL(2, \mathbb{Z})$. This roughly gives us that $SL(2, \mathbb{Z}) = Sp(2, \mathbb{Z})$ is an isomorphism, therefore it has a trivial kernel, and thus $\mathcal{I}(S_g) = 0$ is trivial.

5 Handlebodies and the Torelli group of the Homology 3-sphere

Recall the following definition,

Definition 5.1 (Integral Homology 3-sphere). An Integral Homology 3-sphere S is a 3-manifold with the same Integral Homology as S^3 , that is, $H_1(S) = 0$

We also define the following,

Definition 5.2 (Standard Handle-body). A Standard Handle-body H is a 3-manifold that is locally homeomorphic to closed, regular neighborhood of a graph embedded in a plane in S^3 .

An example of a Standard Handle-body is a solid Torus, which is homeomorphic to a neighborhood of a circle S^1 in S^3 . We can think of Handle-bodies in two ways. Firstly, they are just solid or "filled up" 2-manifolds. Alternatively, we can construct any Handle-body by attaching solid Tori with each other.

Interestingly, the complement of the interior of a Handle-body H in S^3 , H' is itself another Handle-body. From this we can conclude that S^3 is a union of two Handle-bodies with some gluing map $\phi : \partial H \rightarrow \partial H'$,

$$S^3 \cong H \cup_{\phi} H'.$$

The rest of this section dives deeper into why this intuitively makes sense.

Recall that S^n can be identified with $\mathbb{R}^n$ quotiented out by infinity. In particular, this means that we take the the points at infinity of each $\mathbb{R}^n$, and glue/identify them as a singular point. For $\mathbb{R}^1$, this means we glue $+\infty$ and $-\infty$ into a singular point, resulting in a surface that is equivalent to S^1 . For S^2 , we take points at infinity at all directions and glue them into a single point, thus giving us a surface equivalent to a sphere. The analogous statement applies for higher dimensions. This is known as **1-point compactification**, and the illustrations for S^1, S^2, S^3 are given in Fig 3 below.

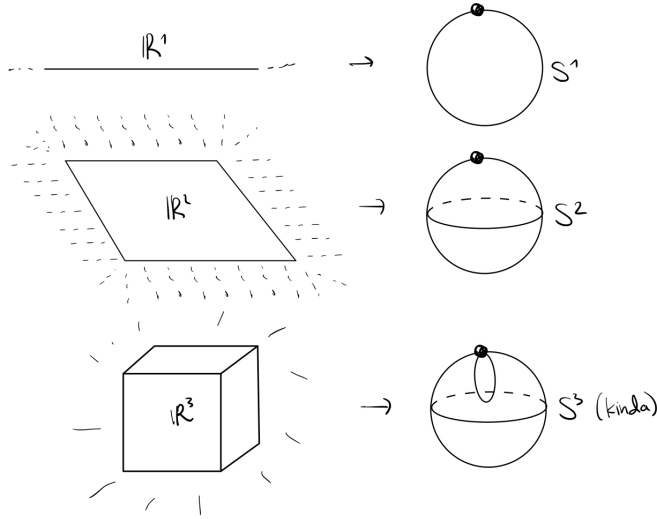


Figure 3: 1-point compactification of S^1, S^2, S^3

Now consider a solid torus $T \cong S^1 \times D^2$ in $\mathbb{R}^3$. and its complement $\text{int}(T)^C$. By thinking of T as a neighborhood of S^1 , we can easily see that we can represent $\text{int}(T)^C$ with a vertical line passing through T .

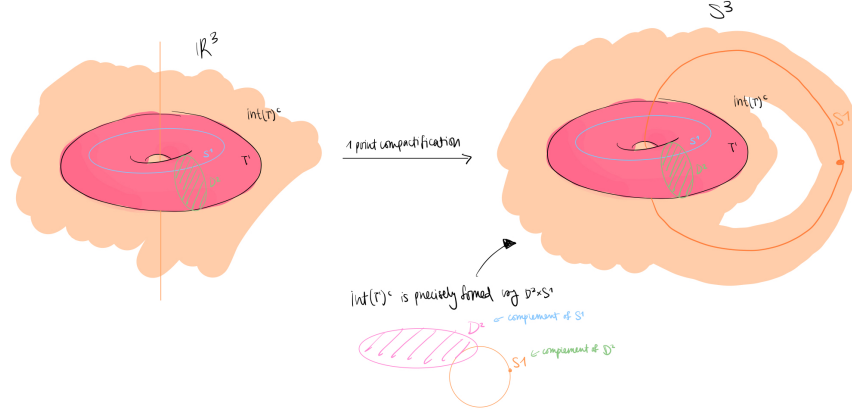


Figure 4: 1-Point Compactification of the Solid Torus Union its Complement.

Applying the 1-point compactification procedure, this vertical line becomes a circle, as illustrated in Fig. 4 below. From this, it is evident that the complement is essentially obtained from the product of said circle and a disc.

In this case, the gluing map is just trivially mapping the boundary of D^2 and S^1 of one solid tori, to S^1 and the boundary of D^2 of the other respectively. Mapping a disk to a circle in this trivial manner effectively "kills" the first homologies of each tori, as we are literally "filling" in the circle (hence the destroying the first homology). Therefore, the first homology of this union is trivial.

We can further generalize this to any handle-body, that is, any solid S_g . In the genus 2 case, we have the illustration in Fig. 5. The blue and purple line going through each of its geni represents the ambient space, and after 1-point compactification, the complement becomes another solid genus 2 tori.

Finally, we discuss the relevance of the Torelli group in Heegaard Splittings.

Definition 5.3 (Heegaard splitting). A Heegaard splitting of a compact, orientable 3-d manifold, M , is the decomposition of M into two Handle-

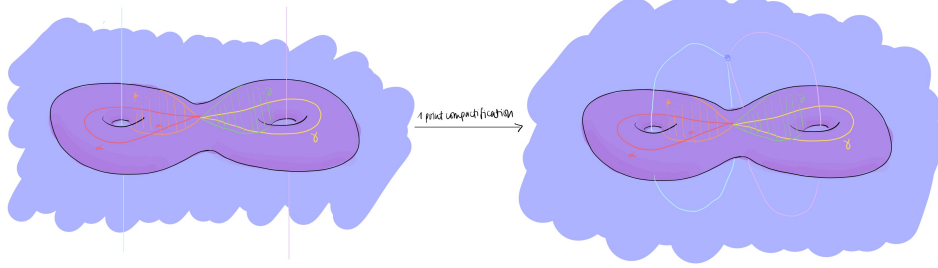


Figure 5: 1-Point Compactification of the Solid S_2 Union its Complement.

bodies with some gluing map ϕ ,

$$M = H \cup_{\phi} H'.$$

Given some self-homologous homeomorphism $\psi : \partial H \rightarrow \partial H$, by pre-composing ψ with the gluing map ϕ , we obtain the 3-manifold,

$$M_{\psi} = H \cup_{\phi \circ \psi} H'.$$

This gives us a way to modify and vary the gluing.

M_{ψ} depends solely on the isotopy class of ψ , thus if $\psi \sim \psi'$, then $M_{\psi} = M_{\psi'}$. However, the homology of this 3-manifold depends on the action of ψ on $H_1(\partial H)$. In particular, $H_1(M_{\psi})$ depends on $\Psi([\psi])$, where Ψ is the symplectic representation from above on $MCG(\partial H)$. Therefore, if $[\psi] \in \mathcal{I}(\partial H)$, then $\Psi([\psi]) = 0$ is trivial, thus M_{ψ} has trivial first homology and is a homology 3-sphere. On the other hand, if $[\psi] \notin \mathcal{I}(\partial H)$, then ψ introduces changes that would not simply "kill" the first homology, thus, $H_1(M_{\psi}) \neq 0$.

In this paper, we introduced Torelli groups as the subgroup of trivial elements of the Mapping Class Group, and as the kernel of the symplectic representation of a surface. We present a few examples of such Torelli group elements, and through this, we saw that the Torelli group is nontrivial and structurally rich for surfaces of genus at least two, in contrast to the torus case where it is trivial. We conclude the discussion with an application to Heegaard splittings, showing that gluing handlebodies via Torelli elements produces integral homology 3-spheres, giving a concrete topological interpretation of Torelli group elements. This illustrates how the Torelli group encodes subtle geometric information that is invisible to homological invariants alone.